

# AnatomyAR

An Augmented Reality anatomy viewer - scan an image, see the 3D organ.

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## 1. What this project is

AnatomyAR is a mobile-friendly web application that helps students learn human anatomy using Augmented Reality (AR). Instead of looking at flat textbook pictures, the student points their phone camera at a printed card and a real 3D model of that organ appears on the screen, floating over the card. The model can be rotated, zoomed and moved to study it from any angle.

It is built from the original project brief, “*AR-Based Interactive Anatomy Learning Platform*”, focused on the single most important feature: **scan an image** → **show the basic 3D model of that organ**.

## 2. What was built

The core request - *scan an image and see its 3D model* - is fully working, and the app was then taken well beyond it:

- **AR Anatomy Viewer** - scan a card with the phone camera and the matching 3D organ appears, anchored on the card. Pinch to zoom, drag to rotate. A live info card shows its name, system and a key fact, and you can snap an **AR photo**.
- **Immersive 3D study pages** - every organ opens in a full interactive viewer with numbered **tap-to-learn hotspots** (parts and what they do), key-fact cards and a fun fact.
- **Voice explanations** - a built-in narrator reads any organ or part aloud, plus a hands-free **guided tour** that rotates the model while explaining each part.
- **AI Anatomy Assistant** - a chat that answers questions about organs and body systems. Works offline out of the box; optionally connect a Claude API key for full AI answers.
- **Quiz mode** - instant multiple-choice quizzes per organ or a mixed quiz, with scoring and feedback.
- **Six real organs**: Heart, Brain, Lungs, Kidney, Pelvis (skeletal) and Liver.
- **Installable app (PWA)** - can be added to the phone's home screen and keeps working offline after the first visit.
- **Fast loading** - the 3D models are compressed (about 7× smaller) so they appear in a second or two.
- **Printable scan cards** - one marker per organ (see Scan\_Cards.pdf).

Deliberately kept out (from the original brief) to stay focused:

- Layered body-system switching (peeling between skeletal / muscular / nervous, etc.).
- User accounts, login and long-term progress tracking / streaks.
- Saved notes, bookmarks and an admin upload panel.

The structure leaves room for these later, but they aren't needed for a strong demo.

## 3. What is in the project folder

| File / folder | What it does                           |
|---------------|----------------------------------------|
| index.html    | Home page + 3D gallery of every organ. |

|                                     |                                                                    |
|-------------------------------------|--------------------------------------------------------------------|
| organ.html                          | Immersive 3D study page (hotspots, facts, voice, guided tour).     |
| scan.html                           | The AR scanner - the camera detects a card and shows the 3D organ. |
| quiz.html                           | Multiple-choice quizzes with scoring.                              |
| markers.html                        | On-screen version of the printable scan cards.                     |
| models/                             | The six 3D organ models (compressed .glb files).                   |
| markers/                            | The barcode marker images used on the cards (0.png-5.png).         |
| css/ , js/                          | Styling and the app code (data, pages, AI assistant, AR).          |
| manifest.webmanifest, sw.js, icons/ | Make it an installable, offline-capable app (PWA).                 |
| Scan_Cards.pdf                      | Printable cards - print this and scan the cards.                   |
| Project_Guide.pdf                   | This document.                                                     |
| README.md                           | The same information in plain text, for developers.                |

## Organ ↔ card mapping

| Card # | Organ  | Body system        |
|--------|--------|--------------------|
| 0      | Heart  | Circulatory System |
| 1      | Brain  | Nervous System     |
| 2      | Lungs  | Respiratory System |
| 3      | Kidney | Urinary System     |
| 4      | Pelvis | Skeletal System    |
| 5      | Liver  | Digestive System   |

## 4. How to put it online (needed for the phone)

**Important:** a phone camera only works on a web page served over **HTTPS** (a browser security rule). Opening the HTML files directly from the folder will *not* let the scanner use the camera. So the folder must be uploaded to a free host that provides an HTTPS link. The easiest option is Netlify Drop - no account or coding needed.

### Option A - Netlify Drop (easiest, ~2 minutes)

- 1 On a laptop, open <https://app.netlify.com/drop> in any browser.
- 2 Drag the whole **IDT** project folder onto the page where it says 'Drag and drop'.
- 3 Wait a few seconds. Netlify gives you a link like <https://random-name-123.netlify.app>.
- 4 Open that link on your phone (type it in, or send it to yourself / show a QR code).
- 5 Tap **Open AR Scanner**, allow the camera, and point it at a printed card.

To update it later, just drag the folder onto Netlify Drop again - you get a fresh link each time, which is fine for a demo.

### Option B - GitHub Pages (free permanent link)

- 1 Create a free GitHub account and a new public repository.
- 2 Upload all the files from the IDT folder into the repository.
- 3 In the repository: **Settings** → **Pages** → **Build from branch** → **main** → **Save**.
- 4 After a minute GitHub gives a permanent <https://<username>.github.io/<repo>/> link.
- 5 Open that link on the phone and use it exactly like Option A.

### Option C - Quick laptop preview only (no camera)

To check the gallery and 3D models on the laptop (not the camera scanner), open a terminal inside the IDT folder and run:

```
python -m http.server 8000
```

Then open <http://localhost:8000> in the browser.

## 5. How to present / demo it

Before the presentation:

- Deploy it (Option A or B above) and confirm the link opens on your phone.
- Print **Scan\_Cards.pdf**, or keep it open on a second screen/laptop to scan from.
- Test each card once so you know the models appear. Good, even lighting helps a lot.
- Charge the phone and connect to reliable Wi-Fi (the models download the first time).

Suggested 3-minute demo flow:

- 1 Explain the problem: anatomy is hard to understand from flat 2D textbook images.
- 2 Open the deployed link on the phone and show the home gallery - tap the **Heart** to open its 3D study page.
- 3 Tap a numbered **hotspot** on the heart to reveal a part and what it does, then press ■ **Listen** or ■ **Guided tour** to let it explain itself out loud.
- 4 Open the **AR Scanner**, point it at the Heart card - the 3D heart appears anchored on the card. Pinch to zoom, drag to rotate, and tap ■ to capture an AR photo.
- 5 Open the **AI Assistant** (■) and ask "what does the liver do?" to show instant answers.
- 6 Open the **Quiz**, answer a couple of questions to show scoring.

- 7 Finish by **installing the app** to the home screen ('Add to Home screen') to show it behaves like a real native app.

Optional: the AI Assistant works offline out of the box. For full conversational answers, tap the ■ link inside the chat and paste a Claude API key (stored only on that phone). Mention possible next steps: layered body-system peeling and saved progress.

## 6. Adding or changing an organ (optional)

- 1 Put a new model file (name .glb) into the **models/** folder.
- 2 Add an entry in `js/organs.js` with the next free card number (copy an existing entry and fill in its name, system, facts, parts and quiz questions).
- 3 Copy one `<a-marker>` block in **scan.html**, change its 'value' to the new card number and its model 'src'.
- 4 Re-run `python make_pdfs.py` to refresh the scan cards.

Models are automatically centred and resized to sit on the card, so they do not need manual scaling.

## 7. Tech stack & credits

| Part                    | Technology                                             |
|-------------------------|--------------------------------------------------------|
| AR engine               | AR.js + A-Frame (marker tracking in the browser)       |
| 3D viewer & study pages | Google <model-viewer> with hotspots                    |
| Voice                   | Browser Web Speech API (text-to-speech)                |
| AI assistant            | Offline knowledge base, optional Claude API            |
| 3D models               | HuBMAP CCF 3D Reference Library (Draco-compressed)     |
| App / offline           | PWA - web manifest + service worker                    |
| Runs on                 | Any modern phone browser (Android Chrome / iOS Safari) |

**Attribution (please keep this):** The 3D organ models are from the HuBMAP Consortium CCF 3D Reference Object Library and are licensed under Creative Commons Attribution 4.0 (CC BY 4.0). Source:  
<https://hubmapconsortium.github.io/ccf/pages/ccf-3d-reference-library.html>